8/10/24

Experiment - 12
End to End Communication at Transport Layer

Aim:
Implement echo client&server using TCP/UDP Sockets.

TCP

```
import socket
import threading
def start_server():
    server = socket.socket(socket.AF_INET, socket.SOCK_
                                        STREAM)
    server.bind(('127.0.0.1', 5000))
    server.listen()
    print("[Server] Listening on 127.0.0.1:5000")
    con, addr = server.accept()
    print(f"[Server] Connected by {addr}")
    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower() == 'quit':
            print("[Server] Connection closed by client.")
            break
        print(f"[Server] Received: {data}")
        conn.send(f"Server recieved: {data}".encode())
    con.close()
    server.close()
def start_client():
    client = socket.socket(socket.AF_INET, socket.
                        SOCK_STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] Connected to Server")
    while True:
        msg = input("Enter message (or 'quit' to exit):")
        client.send(msg.encode())
```

```
        if msg.lower() == 'quit':
            break
        response = client.recv(1024).decode()
        print(f"[Server Response] {response}")
    client.close()
if __name__ == "__main__":
    server_thread = threading.Thread(target=start_server,
        daemon=True)
    server_thread.start()
    start_client()
```

Output:

[Server] Listening on 127.0.0.1:5000 [Client] Connected to Server
[Server] Connected by ('127.0.0.1', 53458) Enter message (or 'quit' to exit):

hello
[Server] Received : hello
[ServerReponse] Server recieved: hello.
Enter message (or 'quit' to exit):
[Server] Listening on 127.0.0.1:5000 [Client]
Conneted to Server
[Server] Connected by ('127.0.0.1', 60328) Enter message (or 'quit to exit):
HI
[Server] Recieved : HI
[Server Response] Server recieved: HI
Enter menage (or 'quit' to exit): quit
[Server] Connection closed by client.

8)

UDP

```
import socket
import threading
def udp_server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] client {addr} ended the
                connection.")
            break
        print(f"[SERVER] Received from {addr}: {msg}")
        s.sendto(data, addr) #Echo back
    s.close()
    print("[SERVER] Stopped.")
def udp_client():
    c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message (or 'quit' to stop):")
        c.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
            print("[CLIENT] closing connection...") break
        data, _ = c.recvfrom(1024)
        print("Echo from server:" data.decode())
    c.close()
    print("[CLIENT] Stopped.")
```

```
if __name__ == "__main__":
    threading.Thread(target=udp_server,
                     daemon=True).start()udp_
                                  client()
```

Output:

```
Enter Message(or 'quit' to stop): [SERVER] Running on
                                  127.0.0.1:12000 hi
[SERVER] Received from ('127.0.0.1', 49670): hi
Echo from Server: hi
Enter message (or 'quit' to stop): hello
[SERVER] Recieved from ('127.0.0.1', 49670): hello
Echo from Server: hello
Enter message (or 'quit' to stop): how are you
Echo from Server: how are you
Enter message (or 'quit' to stop): quit
[CLIENT] closing connection... [SERVER] client ('127.0.0.1
                                  ' 49670) ended the connection.
[CLIENT] Stopped [SERVER] Stopped
```

Result:

Therefore implementation of echo client server Using TCP/UDP Sockets has been executed Successfully.